Scheduling long-haul battery-electric trucks: coordinating charging, breaks, and daily rests

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Abstract

Recent advances in battery technology and fast charging are making long-haul battery-electric freight transport increasingly feasible. As vehicle ranges extend beyond regional distribution, charging decisions become increasingly intertwined with Hours-of-Service regulations. Since charging is time-consuming and unavoidable on long routes, operators must ensure that vehicles remain sufficiently charged while complying with driver regulations. Efficient operation therefore depends on synchronizing charging activities with mandatory driver breaks and daily rest periods. We study this scheduling problem for battery-electric trucks operating on multi-day long-haul routes. We introduce a multi-day electric truck driver scheduling problem in which charging decisions are optimized jointly with Hours-of-Service compliance to minimize total route duration. The problem is formulated as a resource-constrained shortest path problem and solved using an exact labeling algorithm that jointly models battery dynamics and driver regulations. Computational experiments show that optimized battery-electric truck schedules can, under favorable charging conditions, be completed within XX% of the duration of equivalent diesel-truck operations. However, schedule feasibility and performance deteriorate rapidly when charging opportunities are sparse or charging power is insufficient. Compared with a simpler charging strategy, the proposed optimization approach reduces route durations by up to XX%. Moreover, allowing partial recharging provides modest but consistent reductions in route time relative to full recharging. The results demonstrate that synchronization of charging and mandatory driver interruptions is a key operational enabler of long-haul freight electrification.

Keywords: Battery-electric trucks; long-haul freight transport; hours-of-Service regulations; charging scheduling; driver scheduling; resource-constrained shortest path.

1. Introduction

The electrification of road freight transport has progressed rapidly over the last decade. Battery-electric trucks are now being adopted in urban distribution and regional logistics, where vehicles can

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typically recharge at depots or terminals during predictable off-duty periods (Shoman et al., 2023). Recent improvements in battery capacity and fast-charging technology are extending feasible driving ranges and making battery-electric trucks increasingly viable for long-haul freight operations. Current commercial vehicle offerings illustrate this development: heavy-duty battery-electric trucks are commonly marketed with single-charge ranges of around 300 km, while newer long-haul models target approximately 500 km under favorable operating conditions (Volvo Trucks, 2025b; Mercedes-Benz Trucks, 2025). However, road freight transport is a highly competitive business with low profit margins (Council of the European Union, 2025), where schedules are tight and time losses translate directly into cost. If charging causes additional time losses, battery-electric trucks can quickly become unattractive for long-haul freight operations.

A defining feature of long-haul trucking is that operations are constrained by Hours-of-Service (HoS) regulations. Under EU regulation, drivers must take mandatory breaks and daily rest periods, typically taken overnight, that reset driving and working-time limits (Goel & Irnich, 2017). Traditionally, compliance with these regulations has been the primary scheduling challenge, as diesel refueling is fast and widely available. Consequently, refueling is rarely modeled as a binding scheduling resource in conventional truck operations (Schulz, 2024). For battery-electric trucks, in contrast, charging is substantially more time-consuming than diesel refueling and depends on the availability of suitable charging infrastructure. Operators must therefore ensure that vehicles remain sufficiently charged while simultaneously complying with driver regulations. Efficient operation depends on synchronizing charging activities with mandatory breaks and daily rests such that charging can be absorbed into periods during which the driver is already required to be off duty.

In this paper, we investigate the synchronization of charging activities with mandatory breaks and daily rests in long-haul battery-electric freight transport. We study a multi-day scheduling problem in which charging, breaks, and daily rests must be jointly planned to minimize route duration while satisfying both Hours-of-Service and energy constraints.

1.1. Related literature

The Truck Driver Scheduling Problem (TDSP) concerns the construction of driver schedules that satisfy regulatory HoS constraints while optimizing operational objectives such as route duration or transportation cost. The Truck Driver Scheduling Problem (TDSP) has been studied extensively under Hours-of-Service regulations (Archetti & Savelsbergh, 2009; Goel, 2012; Goel & Kok, 2012).

Methodologically, TDSPs are commonly formulated as shortest path problems with resource constraints (SPPRCs). Feasible schedules are constructed incrementally through labeling algorithms that extend partial paths while tracking resources such as accumulated driving and working time since the last break or rest. Resource extension functions are used to update resource states and identify infeasible path extensions (Goel, 2010; Drexl & Prescott-Gagnon, 2010). Building on this paradigm, Goel & Irnich (2017) introduced an exact branch-and-price framework with a parameter-free network representation that avoids time discretization, while preserving regulatory feasibility. However, these models assume conventional vehicles with negligible refueling times and do not consider energy constraints.

While TDSP research focuses on regulatory feasibility, the Electric Vehicle Routing Problem (EVRP) literature provides a rich body of work on modeling battery dynamics and charging decisions (Schneider et al., 2014; Han et al., 2025). Many exact EVRP approaches are based on branch-and-price frameworks in which a master problem selects vehicle routes, while SPPRC pricing problems generate new feasible routes. In these formulations, battery state is treated as a resource and charging duration directly affect route timing (Desaulniers et al., 2016; Lam et al., 2022; Nafstad et al., 2025). Several works allow partial recharging and nonlinear charging functions to better reflect operational practice (Montoya et al., 2017; Froger et al., 2019). Whereas most EVRP studies focus on route construction and charging decisions along a route, recent work has begun to integrate routing decisions with the scheduling of charging infrastructure. For example, Yamín et al. (2025) jointly optimize urban delivery routes and overnight depot-charging schedules using a branch-price-and-cut algorithm. Their formulation accounts for nonlinear charging and limited charging capacity, but does not consider HoS-constrained driver schedules.

Only a limited number of studies explicitly consider the interaction between charging activities and driver breaks and rests under HoS regulation. Schiffer et al. (2017) extend the EVRP by incorporating HoS regulations and allowing en-route charging activities to be synchronized with mandatory driver breaks. Their EVRP is motivated by one-day routing and scheduling of electric commercial delivery vehicles in regional distribution networks. They solve the problem using a mixed-integer formulation combined with an adaptive large neighborhood search heuristic. Closer to the long-haul setting considered in this paper, Wan et al. (2024) study an opportunity-charging planning problem for a battery-electric truck in a single freight corridor case study. Their model coordinates charging decisions with planned stops at highway service areas and logistics hubs while accounting for charging technologies, electricity prices, and temporal flexibility. However, HoS regulations are represented only through simplified driving-time and rest constraints and the model does not consider recurring daily-rest requirements. More recently, Brockmann & Guajardo (2025) develop a mathematical programming model for an Electric Vehicle Truck Driver Scheduling Problem (EVTDSP) in which charging operations and mandatory driver breaks are jointly optimized. Their analysis highlights the importance of accounting for driver regulations when assessing the operational performance of battery-electric trucks. However, the study is restricted to single-day operations and does not consider recurring daily rests or multi-day scheduling decisions. Finally, Schneider et al. (2023) analyze one-day electric truck operations and evaluate predefined charging and EU HoS-complient break strategies, including alternatives enabled by high-power charging technologies such as the Megawatt Charging System. While their study provides valuable insights into the interaction between charging technology and driver regulations, it only evaluates a limited number of predefined operating strategies and does not address the underlying scheduling problem.

1.2. Research gap and contributions

It remains unclear to what extent charging can be absorbed within mandatory driver breaks and daily rests in multi-day long-haul operations, and how incomplete synchronization affects route duration and schedule feasibility. Existing studies focus primarily on distribution networks and single-day operations, whereas

multi-day operations involve repeated charging, mandatory breaks, and recurring daily rests, creating greater potential for both synergies and conflicts between energy and regulatory requirements. Furthermore, existing synchronization studies rely primarily on mixed-integer formulations and heuristic solution approaches and do not provide an exact method for jointly optimizing charging, breaks, and daily rests over multiple days.

To address this gap, we study a multi-day variant of the Electric Vehicle Truck Driver Scheduling Problem (EVTDSP), which determines how charging activities, mandatory breaks, and daily rest periods should be scheduled and synchronized in long-haul battery-electric freight operations. Given a prescribed route and a set of charging opportunities, the objective is to minimize total route duration while satisfying both HoS and battery constraints.

The contributions of this paper are threefold. First, we extend the truck driver scheduling problem to the multi-day battery-electric setting by explicitly integrating charging decisions with both mandatory breaks and recurring daily rest periods. Second, we develop an exact resource-constrained shortest path formulation and labeling algorithm by extending the framework of Goel & Irnich (2017) with battery dynamics, tailored resource extension functions, and dominance rules. Third, we introduce an open-source benchmark set for the EVTDSP informed by interviews and operational data collected from long-distance trucking companies in Norway and Northern Europe, and use it to systematically analyze how route structure, charging conditions, and rest requirements affect schedule feasibility, charging-rest synchronization, and the value of partial recharging.

2. Problem definition

updated based on Magnus his new version. OK Now?

This paper studies the Electric Vehicle Truck Driver Scheduling Problem (EVTDSP), which concerns the construction of HoS-compliant and energy-feasible schedules for a battery-electric truck operating on a long-haul route. Given a predefined sequence of customer visits, the objective is to minimize the total route duration by determining the timing of mandatory breaks and daily rests, as well as when, where and how much to charge the battery. We consider a one-week planning horizon, $T^{\text{horizon}} = 168$ hours, reflecting the regulatory limit on the duration of a weekly work cycle.

The problem is defined on a directed graph $\mathcal{G} = (\mathcal{N}, \mathcal{A})$, where $\mathcal{N}$ and $\mathcal{A}$ denote the sets of nodes and arcs, respectively. The node set comprises the customer nodes $\mathcal{N}^C$, the charging-station nodes $\mathcal{N}^S$, the origin o, and the destination d, i.e., $\mathcal{N} = \mathcal{N}^C \cup \mathcal{N}^S \cup \{o, d\}$. Because the sequence of customer visits is predetermined and only charging stations located along this route are considered, all nodes have a unique order. Let $(i_1, \ldots, i_{|\mathcal{N}|})$ denote this order, with $i_1 = o$ and $i_{|\mathcal{N}|} = d$. The arc set connects consecutive nodes: $\mathcal{A} = \{(i_k, i_{k+1}) \mid k = 1, \ldots, |\mathcal{N}| - 1\}$. Figure 1 illustrates the graph for a route with three customers and six charging stations.

Each customer node $i \in \mathcal{N}^C$ requires a fixed, non-interruptible service time s_i. For each arc $(i, j) \in \mathcal{A}$, the distance d_{ij} and driving time t_{ij} are known. We assume a constant driving speed v and a constant energy-consumption rate α per unit of distance. The vehicle has battery capacity B, departs from the origin

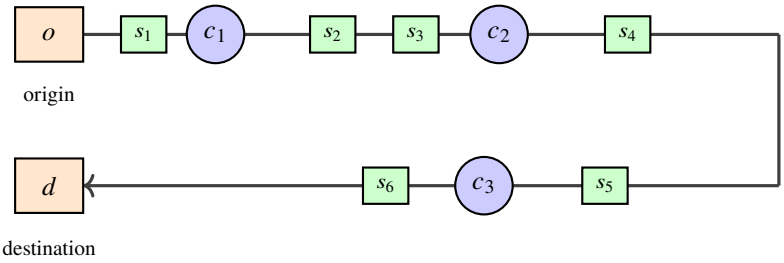


Figure 1: Illustrative route structure for the EVTDSP. The truck travels from origin o to destination d, visiting customers (c_1, c_2, c_3), with charging stations $(s_1, \ldots, s_6)$ available along the route.

fully charged, and must maintain a nonnegative state of charge throughout the route.

All charging stations are assumed to provide a constant charging power P^{ch}. Charging is therefore linear: adding q units of energy requires q/P^{ch} units of time. At a charging station, the vehicle may continue without charging or charge any amount up to the available battery capacity. Partial charging is thus part of the general problem definition, while full charging is obtained by restricting the vehicle to charge to capacity whenever a charging activity is selected. Multiple charging stops are allowed, but charging cannot be performed during customer service.

The driver is subject to the EU HoS rules represented by four limits: $t_{\texttt{drive}|B}$ and $t_{\texttt{elapsed}|B}$ bound the driving and elapsed working time, respectively, between consecutive breaks or rests, while $t_{\texttt{drive}|R}$ and $t_{\texttt{elapsed}|R}$ impose the corresponding limits between consecutive daily rests. A break lasts at least $t_{\texttt{break}}$, and a daily rest lasts at least $t_{\texttt{rest}}$. The driver is fully rested at the origin. Figure 2 illustrates a feasible schedule under these rules. We focus on standard 45-minute breaks and 11-hour daily rests; regulatory options such as split breaks and reduced daily rests are not considered.

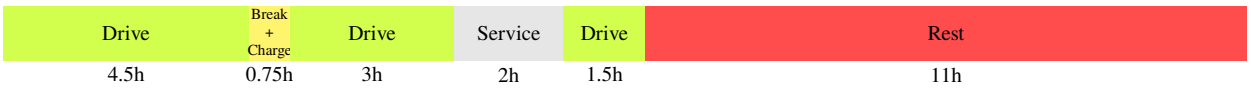


Figure 2: Example of an HoS-compliant schedule for a fully rested driver.

Breaks and daily rests may be taken at customer nodes, charging-station nodes, or intermediate points along an arc, but they cannot overlap with customer service. At charging stations, charging may be performed simultaneously with a break or daily rest. Only the part of a charging activity extending beyond the concurrent break or rest contributes additional time to the schedule.

The EVTDSP therefore determines a sequence of driving, service, charging, break, and rest activities that completes the prescribed route within T^{horizon}, satisfies all HoS and battery constraints, and minimizes the elapsed time from departure at o to arrival at d.

3. SPPRC Framework

- @mostafa+Magnus: resources elapsed since break or rest not interval valued -> Long discussion -> Not needed under our assumptins -> Explain
- @mostafa -> latest|R and latest|B -> only needed for time windows -> remove

The EVTDSP can be formulated as a shortest path problem with resource constraints (SPPRC) and solved using a labeling algorithm. We build on the labeling framework of Goel & Irnich (2017), extending it to jointly represent battery dynamics, charging decisions, and the synchronization of charging with mandatory breaks and daily rests. The prescribed route is transformed into a parameter-free activity network in which driving, breaks, daily rests, customer service, and charging are represented by activity arcs. Feasibility is tracked through temporal, Hours-of-Service, and battery resources propagated using resource extension functions (REFs).

The framework supports both full- and partial-recharge strategies. We focus on the more general partial-recharge formulation, in which the charging amount is a decision variable. The full-recharge formulation is obtained as a special case and is presented in Appendix A.

The main notation used throughout the remainder of the paper is summarized in Table 1. The remainder

Table 1: Main notation.

Symbol	Description
Networks and sets	
$\mathcal{G} = (\mathcal{N}, \mathcal{A})$	Prescribed route graph, with node set $\mathcal{N}$ and arc set $\mathcal{A}$.
$\widetilde{\mathcal{G}} = (\widetilde{\mathcal{N}}, \widetilde{\mathcal{A}})$	Parameter-free activity network used by the labeling algorithm.
$\mathcal{N}^C$, $\mathcal{N}^S \subseteq \mathcal{N}$	Sets of customer and charging-station nodes, respectively.
o, d	Origin and destination nodes.
Route and vehicle parameters	
d_{ij}, t_{ij}	Distance and driving time of route arc $(i, j) \in \mathcal{A}$.
s_i	Service time at customer node i.
v, α	Constant driving speed and energy-consumption rate per unit of distance.
B, P^{ch}	Battery capacity and charging power.
T^{horizon}	Planning horizon.
Hours-of-Service parameters	
t_{break}, t_{rest}	Minimum break and daily-rest durations.
$t_{\texttt{drive}\mid B}$, $t_{\texttt{elapsed}\mid B}$	Driving- and elapsed-time limits between consecutive breaks or rests.
$t_{\texttt{drive}\mid R}$, $t_{\texttt{elapsed}\mid R}$	Driving- and elapsed-time limits between consecutive daily rests.
Labeling notation	
$\mathcal{L} = (i, \pi, \mathbf{T})$	Label at node i, with predecessor information π and resource vector $\mathbf{T}$.
l_r, l'_r	Value of resource r before and after a resource extension.
Δ	Driving duration determined by a driving extension.
q	Energy added during a charging activity.
SOC	Battery state of charge.
f^{start}, f^{drive}, f^{break}, f^{rest}, f^{visit}, f^{charge}	Resource extension functions associated with the activity arcs.

of this section first describes the parameter-free network structure, followed by the label resources, REFs, and the feasibility and dominance rules used in the labeling algorithm.

3.1. Parameter-free activity network

We transform the route graph $\mathcal{G} = (\mathcal{N}, \mathcal{A})$ into a parameter-free network[1] $\mathcal{G} = (\widetilde{\mathcal{N}}, \widetilde{\mathcal{A}})$, on which the labeling algorithm is performed. The two graphs serve different purposes. The route graph $\mathcal{G}$ specifies the ordered sequence of route nodes and the associated travel parameters, whereas $\widetilde{\mathcal{G}}$ represents the feasible sequences of driver and vehicle activities along that route.

The parameter-free network is constructed by replacing each route arc $(i, j) \in \mathcal{A}$ with an activity subnetwork. Its node set $\widetilde{\mathcal{N}}$ consists of the original route nodes and two intermediate state nodes associated with each route arc: a fit state $\tilde{n}_{ij}^{\text{fit}}$ and a dull state $\tilde{n}_{ij}^{\text{dull}}$. Hence,

$$\widetilde{\mathcal{N}} := \mathcal{N} \cup \left\{ \tilde{n}_{ij}^{k} \mid (i, j) \in \mathcal{A},\ k \in \{\text{fit}, \text{dull}\} \right\}.$$

A fit state represents a situation in which driving may continue. A dull state represents a decision point reached after a driving extension. If the current route arc has not yet been completed, a break or daily rest must be taken before driving can resume; if the head node has been reached, the corresponding visit REF may instead be applied.

The arc set $\widetilde{\mathcal{A}}$ contains directed activity arcs connecting the route and intermediate nodes. Each activity arc represents a permissible driver or vehicle activity and is associated with an REF that updates the resources of the label being extended. Figure 3 illustrates the activity subnetworks generated by two consecutive route arcs, (n, s) and (s, m), where $s \in \mathcal{N}^S$ is a charging-station node.

The activity arcs in $\widetilde{\mathcal{A}}$ are: *start arcs*, which initialize the traversal of a route arc; *driving arcs*, which advance the vehicle along the current route arc; *break arcs* and *rest arcs*, which represent the corresponding HoS activities; *visit arcs*, which complete traversal of a route arc and process the route node reached, including customer service where applicable; and *charging arcs*, which determine the charging amount at a charging-station node and initialize traversal of the subsequent route arc. A charging amount of zero represents passing the station without charging.

3.2. Labeling algorithm

We solve the SPPRC using a labeling algorithm on the parameter-free activity network. The algorithm starts with an initial label at the origin o, representing a fully rested driver and a fully charged vehicle. At each iteration, an unprocessed label is extended along every feasible outgoing activity arc. The associated resource extension function (REF) updates the temporal, regulatory, and energy resources. Infeasible and dominated labels are discarded, and a complete schedule is obtained when a label reaches the destination d.

[1]The term *parameter-free* refers to the representation of activity durations. The network does not contain separate arcs for different possible durations of an activity. Instead, the duration is determined by the corresponding REF from the resources of the label being extended. In particular, the driving duration Δ is computed dynamically as the maximum feasible driving increment before the next route node is reached or an HoS limit becomes binding.

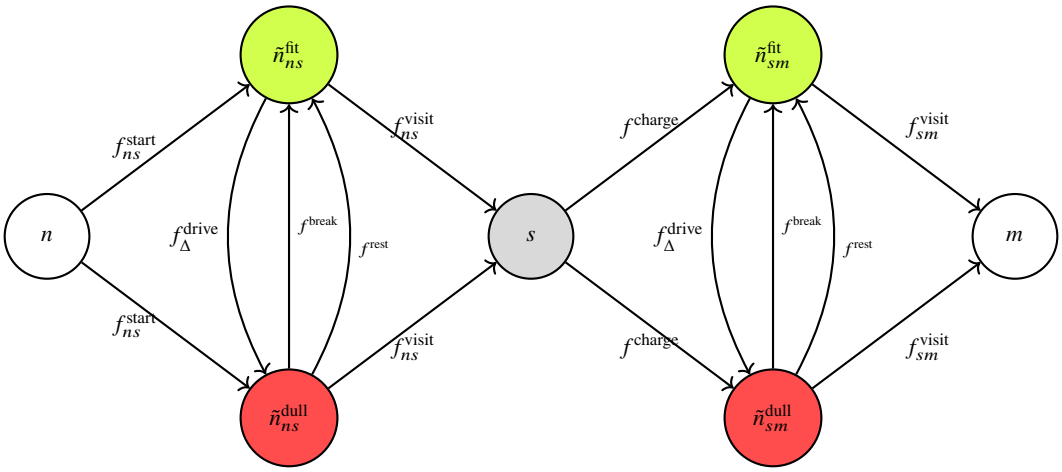


Figure 3: Parameter-free activity subnetworks for two consecutive customer nodes, n and m, with an intermediate charging-station node s. Each activity arc in $\widetilde{\mathcal{A}}$ is associated with the indicated resource extension function.

3.2.1. Label definition

A label represents a partial path in the parameter-free network $\widetilde{\mathcal{G}}$, together with the resource state needed to assess its feasibility and extend it into a complete schedule. Formally, $\mathcal{L} = (i, \pi, \mathbf{T})$, where $i \in \widetilde{\mathcal{N}}$ is the current node, π stores the predecessor information required to reconstruct the partial path, and $\mathbf{T}$ contains the associated resources.

The resource vector $\mathbf{T}$ tracks schedule time, progress along the current route arc, the HoS counters associated with breaks and daily rests, and the vehicle SOC. Table 2 summarizes the individual resources and their feasibility windows. Table 2 summarizes the resources and their feasibility windows.

Table 2: Label resources with corresponding feasibility windows.

Resource	**Type**	**Description**	**Feasibility Window**
`time`	Temporal	Total time elapsed since departure	$[0, T^{\text{horizon}}]$
`timeToNext`	Temporal	Remaining driving time on current route arc	$[0, t_{nm}]$
`drive|R`	Regulatory	Driving time since last daily rest	$[0, t_{\texttt{drive}\|R}]$
`drive|B`	Regulatory	Driving time since last break or rest	$[0, t_{\texttt{drive}\|B}]$
`elapsed|R`	Regulatory	Elapsed time since last daily rest	$[0, t_{\texttt{elapsed}\|R}]$
`elapsed|B`	Regulatory	Elapsed time since last break or rest	$[0, t_{\texttt{elapsed}\|B}]$
`SOC`	Energy	Battery state of charge	$[0,\ B]$

Under partial recharge, the continuous charging decision is represented through interval-valued time and SOC resources, while the HoS resources remain scalar-valued. The interpretation and propagation of these interval endpoints are detailed in Section 3.2.3. The resulting resource vector is

$$\mathbf{T} = \left(l^{\min}_{\texttt{time}}, l^{\max}_{\texttt{time}}, l_{\texttt{timeToNext}}, l_{\texttt{drive}|\texttt{R}}, l_{\texttt{drive}|\texttt{B}}, l_{\texttt{elapsed}|\texttt{R}}, l_{\texttt{elapsed}|\texttt{B}}, l^{\min}_{\texttt{SOC}}, l^{\max}_{\texttt{SOC}}\right).$$

3.2.2. Temporal and HoS resource extensions

Each activity arc $a =\in \widetilde{\mathcal{A}}$ is associated with a resource extension function f_a. Given a resource vector $\mathbf{T}$ at node $i \in \widetilde{\mathcal{N}}$, the REF returns an updated resource vector $\mathbf{T}' = f_a(\mathbf{T})$, computes valid lower bounds on its components, or declares the extension infeasible. Extending a label $\mathcal{L} = (i, \pi, \mathbf{T})$ therefore produces a label $\mathcal{L}' = (j, \pi', \mathbf{T}')$, where π' contains the updated predecessor information. We use l_r and l'_r to denote the value of resource r before and after an extension, respectively.

The activity arcs in $\widetilde{\mathcal{A}}$ use six types of REFs: f_{ij}^{start}, $f_{\Delta}^{\text{drive}}$, f^{break}, f^{rest}, f_{ij}^{visit}, and f_{ij}^{charge}. The start, driving, break, rest, and visit REFs propagate the temporal and HoS resources, while the driving REF additionally accounts for battery-energy consumption. Table 3 summarizes these extensions. For the interval-valued time and SOC resources, the listed updates apply to both endpoints.

Table 3: Resource updates for the start, driving, break, rest, and visit REFs. A dash indicates that the resource remains unchanged.

Resource	f_{ij}^{start}	$f_{\Delta}^{\text{drive}}$	f^{break}	f^{rest}	f_{ij}^{visit}
l'_{time}	—	$l_{\text{time}} + \Delta$	$l_{\text{time}} + t_{\text{break}}$	$l_{\text{time}} + t_{\text{rest}}$	$l_{\text{time}} + s_j$
$l'_{\text{timeToNext}}$	t_{ij}	$l_{\text{timeToNext}} - \Delta$	—	—	—
$l'_{\text{drive\|R}}$	—	$l_{\text{drive\|R}} + \Delta$	—	0	—
$l'_{\text{drive\|B}}$	—	$l_{\text{drive\|B}} + \Delta$	0	0	—
$l'_{\text{elapsed\|R}}$	—	$l_{\text{elapsed\|R}} + \Delta$	$l_{\text{elapsed\|R}} + t_{\text{break}}$	0	$l_{\text{elapsed\|R}} + s_j$
$l'_{\text{elapsed\|B}}$	—	$l_{\text{elapsed\|B}} + \Delta$	0	0	$l_{\text{elapsed\|B}} + s_j$
l'_{SOC}	—	$l_{\text{SOC}} - \alpha v \Delta$	—	—	—

For a route arc $(i, j) \in \mathcal{A}$, t_{ij} denotes the required driving time from route node i to route node j, and s_j denotes the service time at node j (but also index of a charging station), with $s_j = 0$ when $j \notin \mathcal{N}^C$.

The start REF initializes traversal of an arc (i, j) by setting $l'_{\text{timeToNext}} = t_{ij}$. A driving extension advances the schedule by a duration Δ, increases the applicable HoS counters, reduces the remaining driving time, and consumes $\alpha v \Delta$ units of battery energy. The driving duration is determined endogenously as

$$\Delta = \min\Big\{ l_{\text{timeToNext}}, t_{\texttt{drive|}B} - l_{\texttt{drive|B}}, t_{\texttt{drive|}R} - l_{\texttt{drive|R}}, \\ t_{\texttt{elapsed|}B} - l_{\texttt{elapsed|B}}, t_{\texttt{elapsed|}R} - l_{\texttt{elapsed|R}} \Big\}.$$

A driving extension therefore terminates when the head node of the current route arc is reached or when an HoS limit becomes binding. The driving REF additionally updates SOC. The extension is feasible if and only if $l_{\text{SOC}} - \alpha v \Delta \geq 0$.

The break and rest REFs add the corresponding interruption duration and reset the applicable HoS counters. A break resets $l_{\text{drive|B}}$ and $l_{\text{elapsed|B}}$, whereas a daily rest resets both the break- and rest-related resources. Finally, the visit REF completes traversal of route arc (i, j), records arrival at route node j, and adds s_j to the temporal and elapsed-time resources.

3.2.3. Charging resource extension

The charging REF f_{ij}^{charge} introduces the coupling between time and battery energy. We first describe this relationship for a scalar charging decision and then explain how the continuum of charging decisions is represented compactly within a label.

Consider a partial label arriving at charging station i at time t with state of charge l_{SOC}. We assume a linear charging process, such that the energy added is proportional to the charging duration. The vehicle may charge any amount $0 \le q \le B - l_{\text{SOC}}$, requiring a charging duration $\tau(q) = \frac{q}{P^{\text{ch}}}$.

Charging may take place simultaneously with a break or daily rest. In the activity network, this synchronization is represented by applying the corresponding break or rest REF immediately before the charging REF at the same charging-station location. The break or rest REF first adds the interruption duration and resets the relevant HoS resources. The charging REF then recognizes this reset state and treats the already-recorded interruption as a charging window. Thus, the break or rest duration is not added to the schedule a second time.

Let $w(\mathcal{L})$ denote the synchronization window identified from the current HoS state:

$$w(\mathcal{L}) = \begin{cases} t_{\text{rest}}, & \text{if the preceding activity is a daily rest,} \\ t_{\text{break}}, & \text{if the preceding activity is a break,} \\ 0, & \text{otherwise.} \end{cases}$$

Only the portion of the charging duration exceeding this window increases the route duration. The additional, unsynchronized charging time is therefore

$$\delta(q, \mathbf{T}) = \left[\frac{q}{P^{\text{ch}}} - w(\mathbf{T}) \right]^{+}.$$

The charging REF, f_{ij}^{charge}, then updates the time and SOC resources as:

$$l'_{\text{time}} = l_{\text{time}} + \delta(q, \mathbf{T}), \qquad l'_{\text{SOC}} = l_{\text{SOC}} + q, \qquad 0 \le q \le B - l_{\text{SOC}}.$$

The same unsynchronized duration is added to the applicable elapsed-time HoS resources, whereas the accumulated driving-time resources remain unchanged. After charging, `timeToNext` is initialized for the subsequent route arc.

Each feasible charging amount produces a paired departure-time and SOC value. Because charging is linear, the continuum of attainable combinations can be represented by the two endpoints $\left(l_{\text{time}}^{\min}, l_{\text{SOC}}^{\min}\right)$ and $\left(l_{\text{time}}^{\max}, l_{\text{SOC}}^{\max}\right)$. These endpoints define an attainable time–SOC line segment, rather than two independent intervals. The segment represents every feasible partial-charging decision between an earlier departure with less energy and a later departure with more energy.

Charging that fits within an available break or rest window increases SOC without delaying departure.

Consequently, charging less than the amount available at no additional time is not useful. If

$$\frac{B - l_{\mathrm{SOC}}}{P^{\mathrm{ch}}} \leq w(\mathbf{T}),$$

a full recharge can be completed entirely within the synchronized interruption. Otherwise, the label retains the trade-off between departing earlier with less energy and remaining longer to obtain a higher SOC.

The algorithm propagates this trade-off through subsequent driving, HoS, and visit extensions rather than fixing a charging amount immediately. Downstream energy requirements and HoS restrictions determine which portion of the represented interval can lead to a feasible and time-efficient completion. Dominance likewise compares the energy available at relevant times along the represented time–SOC segment. At the destination, the algorithm selects the feasible label with the shortest completion time. By tracing this label backward through the sequence of preceding labels, the algorithm determines the exact charging duration and energy added at each visited charging station.

3.2.4. Feasibility and dominance

clean up! After each resource extension, the resulting label must remain feasible. Feasibility is enforced through the resource windows in Table 2; labels violating any bound are discarded immediately. Under partial recharge, the bounds on `time` and `SOC` apply to both endpoints of the attainable time–SOC segment. The structure of $\widetilde{\mathcal{G}}$ additionally restricts the permissible extensions at each node. Driving arcs may be used only from a fit state. From a dull state, a break or daily rest must be completed before driving resumes. Once $l_{\texttt{timeToNext}} = 0$, a visit arc from either intermediate state completes traversal of the current route arc. At a charging-station route node, charging arcs initialize traversal of the subsequent route arc.

To limit the number of labels stored at each node in $\widetilde{\mathcal{N}}$, dominance is applied among labels ending at the same node. Let $\mathcal{L}_1 = (i, \pi_1, \mathbf{T}_1)$ and $\mathcal{L}_2 = (i, \pi_2, \mathbf{T}_2)$ be two feasible labels at $i \in \widetilde{\mathcal{N}}$. Label $\mathcal{L}_1$ dominates $\mathcal{L}_2$ if and only if:

$$\begin{aligned}
l^{(1)}_{\mathrm{timeToNext}} &\leq l^{(2)}_{\mathrm{timeToNext}}, & & \\
l^{(1)}_{\mathrm{drive|R}} &\leq l^{(2)}_{\mathrm{drive|R}}, & l^{(1)}_{\mathrm{drive|B}} &\leq l^{(2)}_{\mathrm{drive|B}}, \\
l^{(1)}_{\mathrm{elapsed|R}} &\leq l^{(2)}_{\mathrm{elapsed|R}}, & l^{(1)}_{\mathrm{elapsed|B}} &\leq l^{(2)}_{\mathrm{elapsed|B}}, \\
l^{(1)}_{\mathrm{latest|R}} &\geq l^{(2)}_{\mathrm{latest|R}}, & l^{(1)}_{\mathrm{latest|B}} &\geq l^{(2)}_{\mathrm{latest|B}}, \\
l^{\mathrm{min},(1)}_{\mathrm{time}} &\leq l^{\mathrm{min},(2)}_{\mathrm{time}}, & & \\
l^{\mathrm{min},(1)}_{\mathrm{SOC}} &\geq l^{\mathrm{min},(2)}_{\mathrm{SOC}}, & l^{\mathrm{max},(1)}_{\mathrm{SOC}} &\geq l^{\mathrm{max},(2)}_{\mathrm{SOC}}.
\end{aligned}$$

and, at the earliest departure time of label 2, label 1 must offer at least the same battery energy, accounting for optional charging between $l^{\mathrm{min},(1)}_{\mathrm{time}}$ and $l^{\mathrm{max},(1)}_{\mathrm{time}}$. Since the objective is to minimize total route duration, a label is preferable if its resource values are at least as good for continuing the route. For the scalar HoS resources (`timeToNext`, `drive|R`, `drive|B`, `elapsed|R`, and `elapsed|B`), which are monotone

non-decreasing along every REF except break, rest, and synchronized charging, label 1 dominates label 2 at the same node if each of these resources is no larger in label 1 than in label 2. This pairwise comparison is sufficient for SPPRC with monotone resources and preserves optimality (Irnich & Desaulniers, 2005; Goel & Irnich, 2017). Resources `latest|R` and `latest|B` are compared in the reverse direction because larger values correspond to greater remaining scheduling flexibility. The partial-recharge extension introduces interval-valued time and SOC resources. The additional dominance conditions on $l^{\min}_{\text{time}}$, $l^{\min}_{\text{SOC}}$, $l^{\max}_{\text{SOC}}$, and the battery level at $t = l^{\min,(2)}_{\text{time}}$ follow the interval dominance rule for delivery paths with optional recharging (Desaulniers et al., 2016).

4. Experimental design

This section outlines the experimental design used in the computational study. We first describe the benchmark instances used to evaluate the proposed framework. We then introduce a baseline scheduling heuristic, which serves as a reference for assessing the benefits of jointly optimizing charging, breaks, and daily rests.

4.1. Benchmark instances

- SB: create new figure with all customers.

Long-haul freight operations vary substantially in route length, customer distribution, and operating conditions. These characteristics influence the potential gains from jointly scheduling charging, mandatory breaks, and daily rests. We therefore develop a diverse set of structural route instances and evaluate them under different charging-station coverage levels, battery capacities, charging powers, and energy-consumption rates. Rather than reproducing individual observed routes, the instances represent a broad range of realistic long-haul freight operations and constitute, to the best of our knowledge, the first benchmark set for the long-haul EVTDSP.

The benchmark design was informed by interviews with freight operators in Scandinavia and Northern Europe. With assistance from the Norwegian Trucking Association (NLF), 28 companies from different regions and transport segments were contacted, of which eight participated. They represented general freight, container, seafood, temperature-controlled, waste, and bulk transport. The interviews highlighted considerable variation in route length, route duration, number of stops, and customer distribution. Customers were often geographically clustered, resulting in routes that combine long intercity movements with shorter local collection and distribution activities. The interviews also emphasized the importance of charging-station availability and the potential gains from coordinating charging with mandatory driver breaks and daily rests.

Based on these industry inputs, we develop an instance generator for heterogeneous long-haul routes. The generator first defines route instances along three main dimensions: route length, customer distribution, and the number of customer stops. Scenario parameters are added subsequently to vary charging-station

spacing, battery capacity, charging power, and energy consumption without changing the underlying route and customer structure.

Routes are classified as short, medium, or long, corresponding to increasing route distances. Specifically, short routes span 800–1200 km and typically require 1–2 overnight rests, while medium routes span 1500–2500 km and require 3–4 overnight rests. Long routes span 3000–4000 km and require 5–6 overnight rests, reaching the maximum multi-day operating horizon of six days before a longer weekly rest becomes necessary under the HoS regulations. For each instance, the total route length D_{total} is sampled uniformly from the interval associated with its route class.

Each route contains between one and three industrial clusters, interpreted as groups of geographically proximate customers. Cluster centers are positioned along intermediate sections of the route, so that customer visits are concentrated around these clusters and the first and last long-haul legs are typically the longest.

Customer stops are generated in two steps. First, one of three customer-count categories is selected: *few* (1–3 customers), *medium* (4–5 customers), or *many* (6–15 customers), and the total number of customers N is sampled from the corresponding range. Second, customers are allocated among the route clusters, with at least one customer assigned to each cluster. In the base configurations, customers are distributed approximately evenly among clusters, refered to as balanced instances. For selected multi-cluster configurations, an unbalanced variant is also generated by assigning most customers to one randomly selected dominant cluster and only a few to the remaining clusters. For example, 12 customers across three clusters may be allocated as $[4, 4, 4]$ in a balanced instance and $[9, 2, 1]$ in an unbalanced instance.

Customer locations are then placed within 75 km of their cluster center in either direction along the route, while maintaining local customer-to-customer distances of 30–150 km. This produces long intercity movements between clusters and shorter local movements within them. Service times are linked to the customer-count category: 20–30 minutes per stop for few-customer instances, 10–20 minutes for medium instances, and 5–10 minutes for many-customer instances. Instance names indicate the route class, number of clusters, number of customers, and, where applicable, the allocation variant; for example, `long_cl2_cu5_Balanced`.

A total of 43 unique structural benchmark instances are generated by combining the route, cluster, and customer classes described above. Table 4 summarizes the structural design, while the full set of instances and their characteristics is reported in Table D.8. explain why we had to drop 2 instances?

Figure 4 illustrates three representative instances. Charging stations are shown using the base-case maximum spacing of 60 km; the spacing parameter is defined below.

The computational experiments use different technological and infrastructure parameters, as summarized in Table 5.

The selected battery capacities of 350, 500, 750, and 1000 kWh span current and announced heavy-duty electric-truck configurations, which range from approximately 360 to 780 kWh (Winkler et al., 2026; Volvo Trucks, 2025b; Mercedes-Benz Trucks, 2025; Volvo Trucks, 2025a). Charging powers of 350 and 1000 kW represent high-power CCS charging and prospective Megawatt Charging System capabilities, respectively. For example, the Mercedes-Benz eActros 600 supports up to 400 kW with CCS and 1000 kW with MCS

Table 4: Structural design of the benchmark instances.

Route characteristics	**Short**	**Medium**	**Long**
Route length	800–1200 km	1500–2500 km	3000–4000 km
Overnight rests	1–2	3–4	5–6
Sampled customers	2–12	2–13	2–14
Customer characteristic	**Few**	**Medium**	**Many**
Number of customers	1–3	4–5	6–15
Service time per stop	20–30 min	10–20 min	5–10 min
Cluster characteristics			
Clusters per route	1–3 clusters		
Customer distribution across clusters	Balanced; unbalanced		

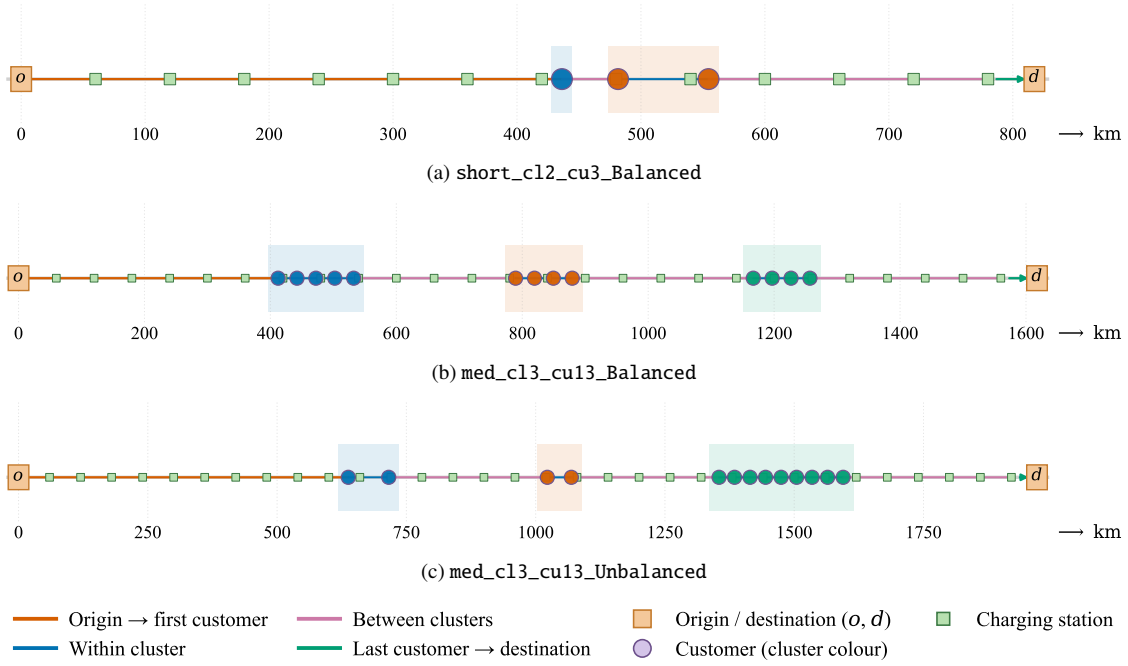


Figure 4: Three representative benchmark instances illustrating variation in route length and customer distribution. Charging stations are shown using a maximum spacing of 60 km. colors of some arcs are a bit weird -> check steffen

Table 5: Scenario parameters and fixed HoS parameters.

Scenario parameters			
Parameter	**Symbol**	**Base value**	**Sensitivity values**
Battery capacity	B	750 kWh	**350**, **500**, 1000 kWh
Charging power	P_{ch}	350 kW	1000 kW
Maximum charging-station spacing	d_{CS}	60 km	40, 80, 100 km
Energy consumption	α	1.49 kWh/km	1.12, 2.08 kWh/km
Fixed HoS parameters			
Parameter	**Symbol**	**Value**	
Maximum continuous driving time	$t_{\text{drive}\|B}$	4.5 h	
Daily driving limit	$t_{\text{drive}\|R}$	9 h	
Maximum elapsed time before break	$t_{\text{elapsed}\|B}$	6 h	
Maximum elapsed time before daily rest	$t_{\text{elapsed}\|R}$	13 h	
Minimum break duration	t_{break}	0.75 h	
Minimum daily rest duration	t_{rest}	11 h	

(Mercedes-Benz Trucks, 2025). In our model, these values are treated as constant charging rates; the 1000 kW case should therefore be interpreted as an idealized near-future scenario. Energy-consumption rates of 1.12, 1.49, and 2.08 kWh/km represent operating conditions ranging from favorable to demanding, around values reported in empirical studies and real-world trials (Volvo Trucks, 2025b; Le Corguillé et al., 2025). Travel times are calculated using a fixed average speed of 60 km/h, based on historical data provided by an industry partner.

The maximum charging-station spacing ranges from 40 to 100 km. This range reflects the 2030 targets in EU Regulation 2023/1804 (AFIR): publicly accessible charging sites for heavy-duty vehicles must be available in each direction of travel at intervals of at most 60 km along the TEN-T core network and 100 km along the comprehensive network; including at least two 350 kW charging points on the core network and one on the comprehensive network (European Parliament and Council of the European Union, 2023).

Charging stations are inserted separately along each route leg. If a leg is shorter than the selected maximum spacing, one station is placed at its midpoint. Otherwise, stations are distributed such that the distance between consecutive charging opportunities does not exceed the selected spacing. The spacing parameter thus represents an upper bound rather than the exact distance between neighboring stations.

The HoS parameters are fixed across all experiments. Following Goel & Irnich (2017), we use maximum continuous and daily driving times of 4.5 and 9 h, maximum elapsed times of 6 and 13 h before a break and daily rest, and minimum break and daily-rest durations of 0.75 and 11 h, respectively.

4.2. Driver-inspired heuristic

To assess how much of the benefit of joint optimization can be achieved with simple operational rules, we develop a driver-inspired sequential scheduling heuristic. It constructs a feasible schedule along the

prescribed route using the current vehicle and hours-of-service (HoS) state and information about the next charging opportunity. Its decisions follow four rules:

1. *Drive as far as reasonably possible.* At a charging station, continue without charging if the available state of charge (SOC) is sufficient to reach the next charging opportunity or the destination, unless a break or daily rest will become mandatory soon and can be synchronized there (Rule 2).
2. *Use a necessary charging stop for a break or rest.* If charging is needed to reach the next charging opportunity, combine the charging stop with a break or daily rest. Take a daily rest if one will become mandatory within t^{sync}; otherwise, take a break.
3. *Take an upcoming break or rest at a charging station.* If a break or daily rest will become mandatory within t^{sync}, take it early at the current charging station and synchronize it with charging. If both interruptions are eligible, prefer the daily rest.
4. *Take an interruption on the road if needed.* If the vehicle is travelling between charging stations and cannot reach the next station before an HoS limit becomes binding, take the required interruption on the road.

The synchronization horizon t^{sync} determines how early an upcoming interruption may be taken at a charging station and whether a daily rest is close enough to take instead of a break during necessary charging. Unless otherwise stated, we use $t^{\text{sync}} = 1$ h.

We evaluate a *full-charge policy*, which charges to full capacity at a planned charging stop, and an *overlap-only policy*, which charges during the concurrent interruption and extends charging only as needed to reach the next charging opportunity. The complete pseudocode is provided in Appendix **??**.

5. Computational results

All algorithms and experimental procedures were implemented in Python [version: Mostafa]. The experiments were executed on [processor model and clock speed], with [RAM], running [operating system:]; [number of cores per run and parallel execution setup]. The labeling algorithms are designed to solve the scheduling problem exactly through label extension, feasibility checks, and dominance. A run is regarded as solved to optimality only when label processing is completed; a feasible solution returned before completion does not establish optimality. [Runtime limit and termination status: all solved to optimality?] The implementation, benchmark instances, and experiment configurations are openly available at [public repository URL].

We first examine the conditions under which optimized electric operation can approach diesel route durations. We then isolate the value of partial rather than full charging and assess the performance lost when joint optimization is replaced by simple scheduling rules.

5.1. Conditions for effective electric operation

We compare the partial-charging labeling solution with a diesel reference for the same route. In the available experiments, the diesel reference is implemented by using a sufficiently large battery capacity

to make energy availability nonbinding, while retaining the driver-scheduling constraints. For route i and electric-vehicle parameter setting θ, the paired route-time penalty is

$$g_{i,\theta} = 100\frac{T^{\mathrm{EV}}_{i,\theta} - T^{\mathrm{D}}_{i}}{T^{\mathrm{D}}_{i}}. \tag{1}$$

We also report the absolute difference in minutes. Percentage gaps are calculated for each route before aggregation, so every route receives equal weight. Feasibility and result availability are considered separately from route duration.

Draft data note: the figures and table below have been regenerated from the inherited partial-charging results. Their saved base case is 750 kWh, 350 kW, 60 km station spacing, and 1.49 kWh/km. This differs from the 500 kWh and 50 km base case in the experimental-design table; agree on the final settings before publication. The available files do not establish complete experiment coverage, original input hashes, or search termination. Missing records must not be interpreted as infeasible instances.

Base-case performance.. Table 6 summarizes the 43 available paired records out of the 45 benchmark routes. The missing records concern two short routes; all 43 available electric and diesel records are reported feasible. Across these pairs, the mean route-time penalty is 0.71%, the median is 0.59%, and the mean additional duration is 37.5 minutes. Of the 43 routes, 38 have a penalty of at most 1%, and 42 have a penalty of at most 5%. These proportions describe the available records rather than the complete benchmark.

Table 6: Draft base-case comparison of partial-charging electric schedules with the diesel reference.

Route class	Pairs	Diesel (h)	Electric (h)	Mean gap (%)	P90 gap (%)	Mean gap (min)	Max. gap (min)
Short	13/15	34.00	34.12	0.40	0.84	7.4	35.5
Medium	15/15	73.13	73.67	0.82	0.93	32.2	180.0
Long	15/15	130.47	131.62	0.87	1.52	69.0	153.5
All routes	43/45	81.30	81.93	0.71	1.20	37.5	180.0

Pairs reports available paired records relative to the 45-route benchmark. All available pairs are reported feasible; missing records have unknown status. Durations and percentage gaps are averaged over routes, with each route weighted equally. P90 is the empirical 90th percentile of paired percentage gaps (linear interpolation).
Saved base settings: 750 kWh, 350 kW, 60 km station spacing, and 1.49 kWh/km. Historical input hashes and solver termination status are unavailable.

Figure 5 shows the route-level variation behind these averages. For short routes, the median penalty is zero, whereas long routes incur a mean additional duration of approximately 69 minutes despite a mean relative penalty below 1%. The largest observed penalty is 5.68%, or 180 minutes, on a medium route. Reporting both minutes and percentages therefore distinguishes a small relative penalty from a potentially consequential increase in completion time. Explaining the exceptional cases requires examining their schedules; the aggregate differences alone do not establish a synchronization mechanism.

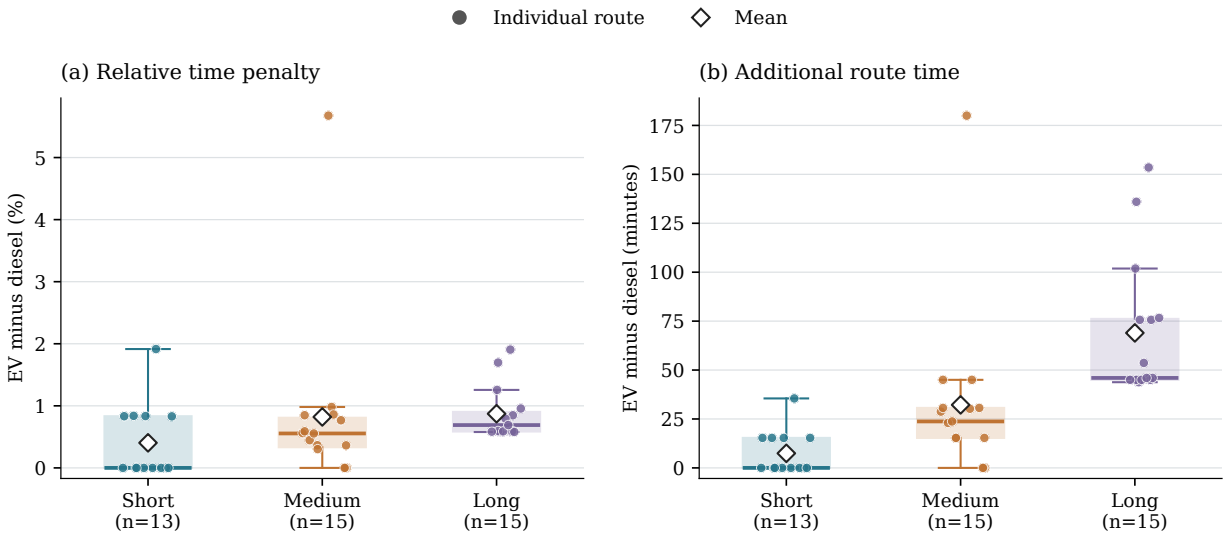
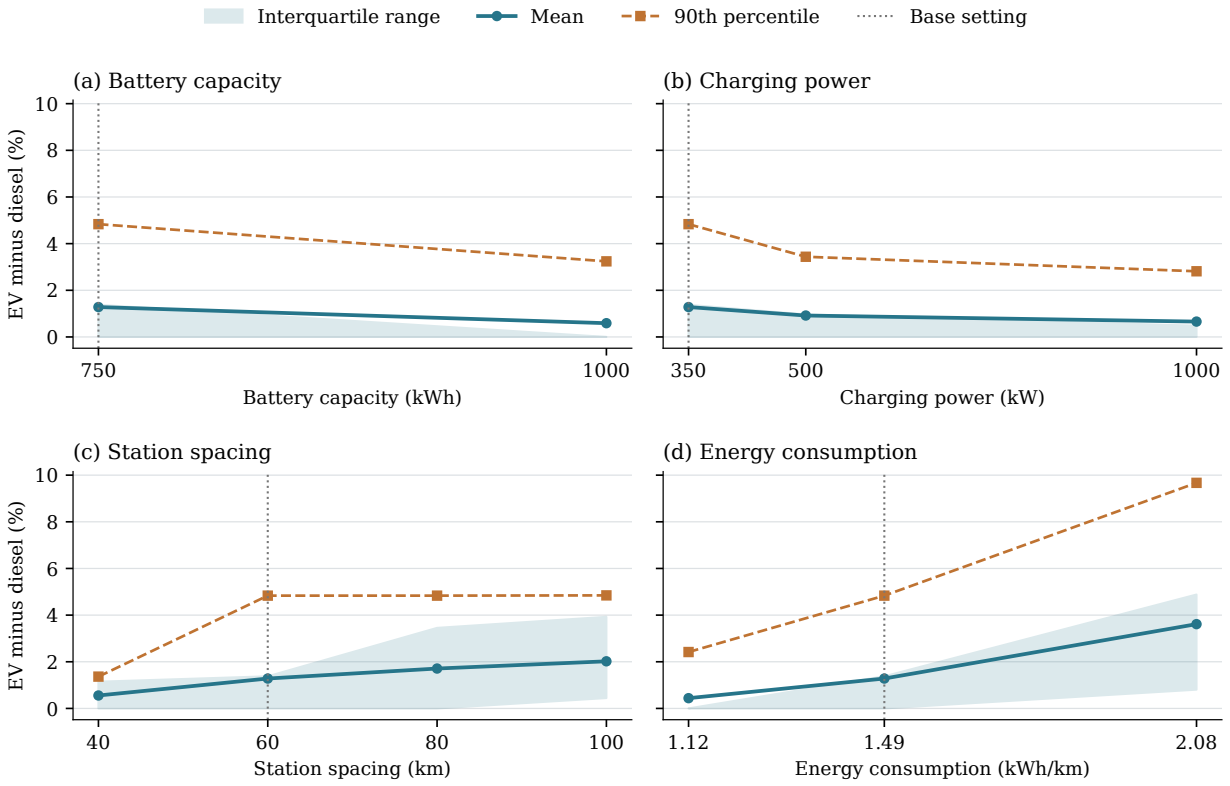


Figure 5: Draft base-case route-time penalties of partial-charging electric schedules relative to diesel, for 43 available route pairs. Dots represent individual routes and diamonds indicate means. Boxes span the interquartile range, with the median shown inside; whiskers extend to the most extreme observations within 1.5 interquartile ranges. All observations, including those outside the whiskers, are shown. The two panels report the same pairs in percentages and minutes.

Technology and infrastructure sensitivities.. The inherited partial-charging sensitivity archive contains 1,273 records across 36 routes, with incomplete coverage of the parameter combinations. To avoid attributing changes in sample composition to parameter effects, Figure 6 uses the same 11 routes at every plotted setting, varying one parameter at a time and holding the others at their saved base values. All 11 routes are short routes and have recorded feasible solutions at every plotted setting. These complete-case comparisons are exploratory and may be selected by solve difficulty. They do not establish sensitivity patterns for medium or long routes, nor a feasibility rate for the intended experiment.

Within this fixed subset, the mean electric–diesel penalty at the base setting is 1.28%. Increasing battery capacity from 750 to 1,000 kWh reduces it to 0.59%, while increasing charging power from 350 to 1,000 kW reduces it to 0.66%. Raising energy consumption from 1.12 to 2.08 kWh/km increases the mean penalty from 0.44% to 3.61%, with a 90th-percentile penalty of 9.67% at the higher consumption level. These observations motivate examining demanding operating conditions as well as averages. The subset differs from the base-case sample in Table 6, so their aggregate penalties should not be compared as a parameter effect.

For station spacing, the mean penalty increases from 0.55% at 40 km to 2.02% at 100 km in this subset. Changing the spacing also changes candidate station locations, so this comparison does not isolate the effect of removing stations from a fixed, nested network. The available extended-spacing results use full charging and are excluded from this partial-charging comparison. Additional partial-charging runs are required to investigate where sparse infrastructure leads to infeasibility.



Exploratory subset: the same 11 short routes at every setting; other parameters held at base values.

Figure 6: Exploratory one-at-a-time sensitivity of the paired electric–diesel route-time penalty, using the same 11 short routes in all panels. Solid lines show means, dashed lines the empirical 90th percentile, and shading the interquartile range; these are descriptive distributions, not confidence intervals. Dotted vertical lines indicate the saved base setting. Lines connect tested settings only. Other parameters remain at their base values.

Interaction between battery capacity and charging power.. Figure 7 crosses battery capacity and charging power at three energy-consumption levels, with station spacing fixed at 60 km and the same 11 short routes in every cell. Each cell reports a mean paired penalty relative to diesel. The figure makes it possible to assess whether the effect of one technology parameter depends on the other, rather than averaging over differing parameter combinations. In the observed subset, penalties are small under favorable consumption, while the absolute benefits of additional battery capacity and charging power are larger at high consumption. This is an initial interaction analysis over the available 750–1,000 kWh battery range; smaller batteries and the other route classes remain to be evaluated.

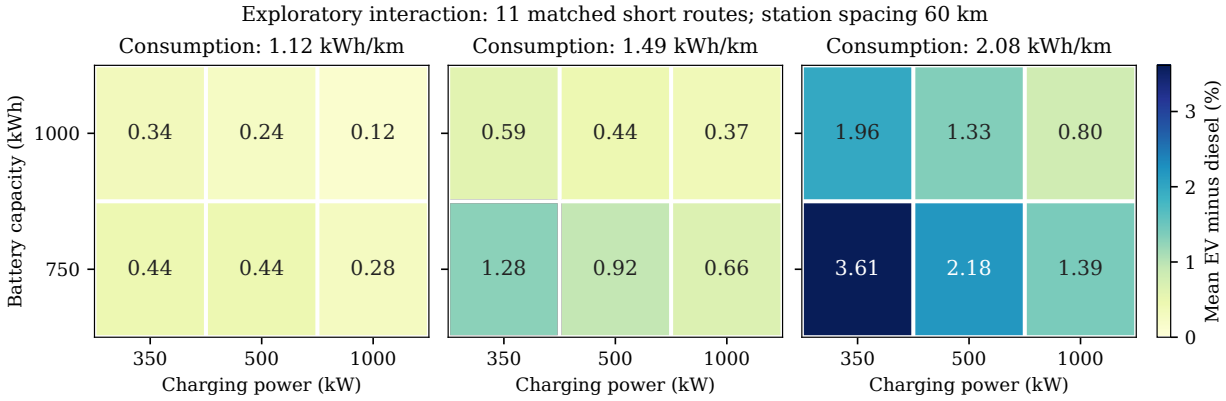


Figure 7: Exploratory battery-capacity–charging-power interaction for 11 matched short routes. Cell values and the common color scale show the mean paired electric–diesel route-time penalty (%). Station spacing is 60 km throughout. The outlined cell marks the saved base configuration. All cells use the same routes and have recorded feasible solutions; missing scenarios elsewhere in the archive have unknown status.

5.2. The value of partial versus full charging

- **Research question.** How much does charging flexibility improve optimized schedules, and when does it matter most?
- **Controlled comparison.** Compare optimized partial and full charging under identical route and scenario conditions. Report the distribution of paired gains and any differences in feasibility.
- **Typical and difficult cases.** Test whether modest typical gains conceal substantial benefits for particular routes or conditions; do not assume this pattern in advance.
- **Mechanisms and targeted sensitivities.** Explain larger gains through charging beyond concurrent interruptions, changes in charging-stop choices, or avoided additional daily rests. Include only sensitivities that clarify the value of partial charging, rather than repeating the full technology study.
- **Planned figures.** One compact paired comparison and an illustrative schedule comparison explaining a substantial gain.

5.3. Simple scheduling rules versus joint optimization

- **Research question.** How much performance is lost when charging, breaks, and rests follow simple rules representing what a driver might do, rather than a jointly optimized schedule?
- **Heuristic comparison.** Evaluate the driver-inspired sequential heuristic with both full-charge and overlap-only policies against optimized partial charging. These are stylized operational rules, not an empirically validated model of driver behavior.
- **Typical losses and difficult cases.** Report the distribution of paired completion-time gaps, including larger losses, and whether the heuristic finds a feasible schedule when the optimized method does.
- **Synchronization horizon.** Examine sensitivity to t^{sync} for both heuristic policies, showing typical and upper-tail losses as well as feasibility outcomes. Explain missed synchronization opportunities and unnecessarily early interruptions.
- **Charging-policy effects.** Identify when overlap-only charging helps and when charging more at an earlier stop avoids later problems. Compare the full-charge heuristic with optimized full charging to distinguish scheduling losses from the full-charge restriction.
- **Planned figures.** A distribution of heuristic performance gaps and a synchronization-horizon sensitivity figure for both policies; use selected schedule examples to explain difficult cases.

The sensitivity analysis in Figure 8 shows a U-shaped pattern in mean route duration for both charging policies. Route duration is minimized near $t^{\text{sync}} = 0.75$ h and remains comparatively low at 1 hour. Shorter horizons miss useful opportunities to synchronize charging and interruptions, whereas longer horizons can trigger interruptions prematurely. We therefore use $t^{\text{sync}} = 1$ h as the default.

6. Conclusion

This paper studied the joint scheduling of charging, mandatory breaks, and daily rests for battery-electric trucks operating on fixed multi-day long-haul routes. We introduced the electric vehicle truck driver scheduling problem as a multi-day extension of the truck driver scheduling setting in which energy replenishment becomes a binding operational resource, and we developed an exact solution approach by extending the parameter-free labeling framework of Goel and Irnich with battery dynamics and charging decisions. The resulting SPPRC formulation captures the interaction between Hours-of-Service compliance and energy feasibility without time discretization, and supports both full- and partial-recharge strategies within a unified framework.

The computational results show that the central operational issue is not charging per se, but charging *in synchrony* with legally mandated interruptions. In the base-case benchmark, partial recharge systematically improves on full recharge, while both exact methods substantially outperform the greedy benchmark. The absolute value of partial recharge grows with route length: although the percentage gain over full recharge

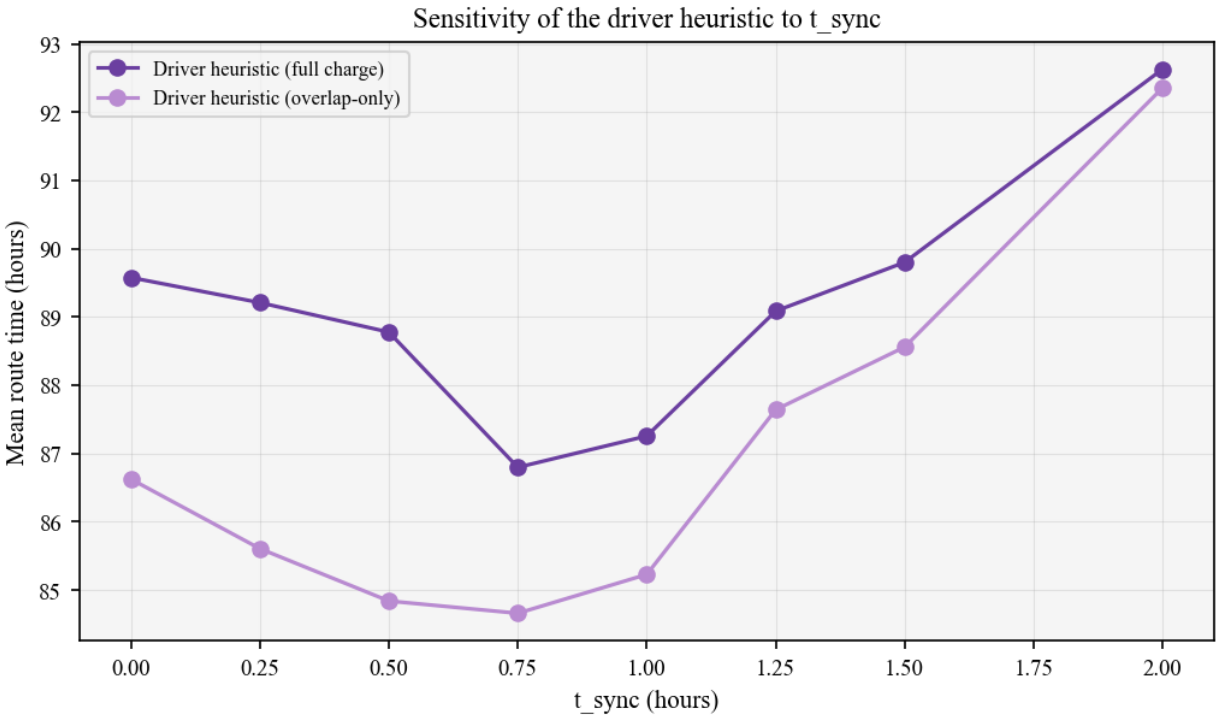


Figure 8: Sensitivity of mean route duration to the synchronization horizon t^{sync} for the full-charge and overlap-only driver heuristics. Horizontal lines indicate the mean route durations of the benchmark policies, and the vertical dotted line marks the default setting $t^{\text{sync}} = 1$ h.

remains modest, the time saving rises from about 12 minutes on short routes to about 49 minutes on long routes. Mechanism-based comparisons show that these gains arise because partial recharge allocates charging more flexibly across stops, increases the share of charging synchronized with breaks and rests, and reduces idle and unsynchronized charging on the routes where such effects matter most. Near-diesel performance is achieved precisely in those instances where charging can be fully absorbed into the Hours-of-Service structure.

These findings carry two broader implications. First, the feasibility and competitiveness of long-haul electric trucking depend critically on whether charging opportunities can be embedded within the driver's regulatory timetable rather than added as separate time-consuming stops. Second, the value of partial recharge is highest in technologically tight settings, namely when energy consumption is high, battery capacity is smaller, or charging power is lower. In such regimes, the scheduling problem becomes more constrained, and the ability to tailor charging amounts to downstream breaks and rests becomes materially more valuable. From a planning perspective, this suggests that operational synchronization should be treated as a first-order design principle alongside battery size and charger deployment.

Several extensions remain relevant for future research. The present study considers fixed route sequences, homogeneous charging technology, deterministic travel and energy consumption. Extending the framework to integrated routing and scheduling decisions, heterogeneous and potentially congested charg-

ing infrastructure, nonlinear charging functions, and stochastic travel-time or energy-consumption processes would further strengthen its practical applicability.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI tools in order to suggest alternative formulations for individual sentences. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix A. Full-recharge SPPRC variant and pass-by arcs

needs to be updated validates, after finishing the partial charging description in the main text

This appendix presents the full-recharge variant of the SPPRC framework. The full-recharge strategy is recovered from the partial-recharge model by restricting every charging action to restore the battery to full capacity. Under this restriction, the energy state can be represented by a scalar resource, and additional pass-by arcs are used to preserve feasible continuations when a charging station is traversed without stopping.

Appendix A.1. Network structure with pass-by arcs

Figure A.9 illustrates the extended subnetwork for a route segment from customer n to customer m with a single intermediate charging station s. The corresponding arcs are (n, s) and (s, m).

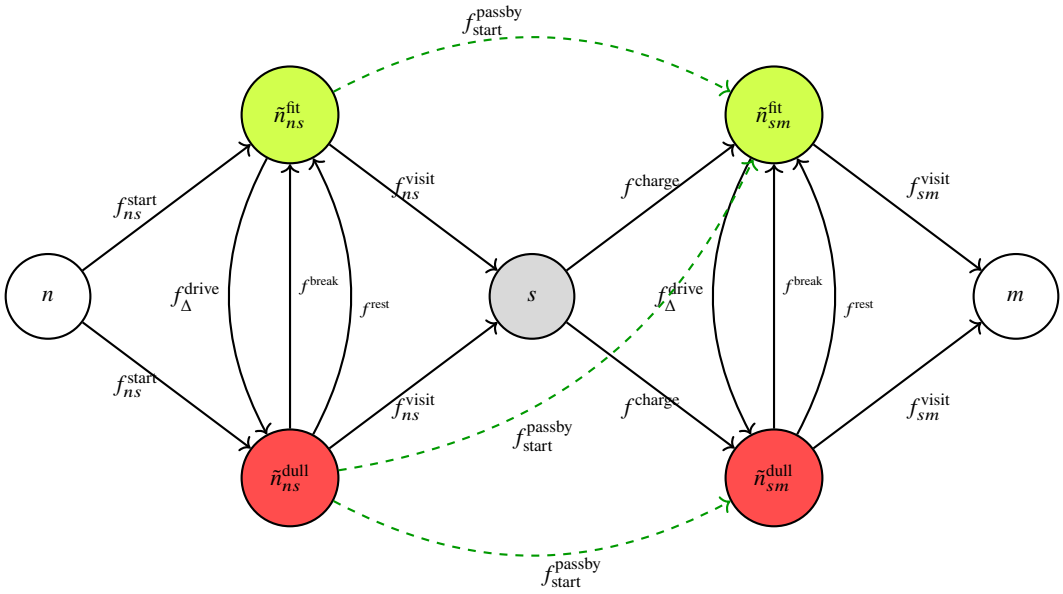


Figure A.9: Extended subnetwork with an intermediate charging station s and pass-by arcs under the full-recharge strategy.

At a charging-station node, the schedule may either initiate a charging activity or continue without charging. Charging is represented through charging arcs originating from the charging-station node, while pass-by arcs allow the station to be traversed without stopping. Together, these arcs preserve the feasible continuation of both fit and dull driver states while introducing the operational choice of whether and where to recharge.

The pass-by arcs determine how HoS states are propagated when a charging station is traversed without stopping. The fit-to-fit pass-by arc represents a driver who reaches the charging-station location in a state that permits further driving, either because no interruption is due or because the required break or rest has already been completed. If the station is skipped, the driver therefore remains fit and can immediately begin the arc from the charging-station location to m. A fit-to-dull pass-by arc is unnecessary, since it would require another break or rest before any additional driving; the dull state is reached naturally only when a

driving limit becomes binding on the subsequent arc. The dull-to-dull pass-by arc preserves an outstanding interruption requirement: after skipping the charging station, the driver remains dull and must take a break or rest before driving toward m. The dull-to-fit pass-by arc is a network connector that preserves the alternative continuations available after the charging-station node, allowing the next arc to begin with driving and, if necessary, a break or rest to be taken later. It does not reset any HoS resource, so such a continuation is retained only when it is feasible under the current driving and working-time counters.

Appendix A.2. Full-recharge label definition

Under the full-recharge strategy, each charging stop restores the battery to full capacity. The energy state can therefore be represented by a single scalar resource l_{reqEC}, which measures the energy requirement accumulated since the last recharge. Equivalently, the state of charge after any partial path is uniquely determined by $B - l_{\text{reqEC}}$, where B denotes the battery capacity.

A full-recharge label is thus defined as

$$\begin{aligned} l = (&l_{\text{time}},\ l_{\text{timeToNext}},\ l_{\text{drive|R}},\\ &l_{\text{drive|B}},\ l_{\text{elapsed|R}},\ l_{\text{elapsed|B}},\\ &l_{\text{reqEC}}) \end{aligned}$$

The resource l_{reqEC} is bounded by $0 \le l_{\text{reqEC}} \le B$. Its design is inspired by Desaulniers et al. (2016), but we retain an energy-based rather than time-based representation in order to preserve flexibility across different battery capacities, charging powers, and energy-consumption rates.

Appendix A.3. Full-recharge resource extension functions

For the full-recharge strategy, charging at a station cs is represented by an additional REF $f_{\text{cs}}^{\text{charge}}$. This extension resets the accumulated energy requirement l_{reqEC} to zero and increases the elapsed time by the charging duration implied by the current energy deficit and the available charging power. When charging overlaps with a legally admissible break or rest, the corresponding HoS counters are updated consistently so that synchronization between charging and regulatory interruptions is captured directly in the label propagation. In addition, pass-by arcs are modeled through $f_{\text{start}}^{\text{passby}}$, which allows a charging station to be traversed without stopping while preserving all resource values except the remaining travel time to the next route node. Table A.7 summarizes the resource updates for the REFs together with the full-recharge charging and pass-by extensions.

Appendix A.4. Full-recharge feasibility and dominance

Under the full-recharge strategy, energy feasibility is captured through the accumulated energy requirement since the last recharge. The scalar resource l_{reqEC} must remain within $[0, B]$ and is reset to zero by a recharge extension.

Table A.7: Resource updates for the REFs and the full-recharge charging extension.

Resource	f_{nm}^{start}	$f_{\Delta}^{\text{drive}}$	$f_{\Delta}^{\text{break}}$	f_{Δ}^{rest}	f_{nm}^{visit}	$f_{\text{cs}}^{\text{charge}}$		$f_{\text{start}}^{\text{passby}}$
l'_{time}	—	$l_{\text{time}} + \Delta$	$l_{\text{time}} + t_{\text{break}}$	$l_{\text{time}} + t_{\text{rest}}$	$l_{\text{time}} + s_m$	$\begin{cases} l_{\text{time}} \\ l_{\text{time}} + \max\left\{0, \frac{l_{\text{reqEC}}}{P^{\text{ch}}} - t_{\text{break}}\right\} \\ l_{\text{time}} + \frac{l_{\text{reqEC}}}{P^{\text{ch}}} \end{cases}$	$\begin{array}{l} l_{\text{elapsed}\|\text{R}} = 0 \\ l_{\text{elapsed}\|\text{B}} = 0 \\ \text{otherwise} \end{array}$	—
$l'_{\text{timeToNext}}$	t_{nm}	$l_{\text{timeToNext}} - \Delta$	—	—	—	$t_{\text{cs},m}$		$t_{\text{cs},m}$
$l'_{\text{drive}\|\text{R}}$	—	$l_{\text{drive}\|\text{R}} + \Delta$	—	0	—	—		—
$l'_{\text{drive}\|\text{B}}$	—	$l_{\text{drive}\|\text{B}} + \Delta$	0	0	—	—		—
$l'_{\text{elapsed}\|\text{R}}$	—	$l_{\text{elapsed}\|\text{R}} + \Delta$	$l_{\text{elapsed}\|\text{R}} + t_{\text{break}}$	0	$l_{\text{elapsed}\|\text{R}} + s_m$	$\begin{cases} 0 \\ l_{\text{elapsed}\|\text{R}} + \frac{l_{\text{reqEC}}}{P^{\text{ch}}} \end{cases}$	$\begin{array}{l} l_{\text{elapsed}\|\text{R}} = 0 \\ \text{otherwise} \end{array}$	—
$l'_{\text{elapsed}\|\text{B}}$	—	$l_{\text{elapsed}\|\text{B}} + \Delta$	0	0	$l_{\text{elapsed}\|\text{B}} + s_m$	$\begin{cases} 0 \\ l_{\text{elapsed}\|\text{B}} + \frac{l_{\text{reqEC}}}{P^{\text{ch}}} \end{cases}$	$\begin{array}{l} l_{\text{elapsed}\|\text{B}} = 0 \\ \text{otherwise} \end{array}$	—
l'_{reqEC}	—	$l_{\text{reqEC}} + \Delta \cdot \alpha$	—	—	—	0		—

Let l_1 and l_2 be two labels ending at the same node. Under the full-recharge strategy, l_1 dominates l_2 if

$$\begin{aligned} &l^{(1)}_{\text{time}} \le l^{(2)}_{\text{time}}, \qquad l^{(1)}_{\text{timeToNext}} \le l^{(2)}_{\text{timeToNext}}, \\ &\mathbf{l}^{(1)}_{\text{drive}} \le \mathbf{l}^{(2)}_{\text{drive}}, \qquad \mathbf{l}^{(1)}_{\text{elapsed}} \le \mathbf{l}^{(2)}_{\text{elapsed}}, \\ &l^{(1)}_{\text{reqEC}} \le l^{(2)}_{\text{reqEC}}. \end{aligned}$$

where

$$\mathbf{l}_{\text{drive}} = \left(l_{\text{drive}|\text{R}},\, l_{\text{drive}|\text{B}}\right), \qquad \mathbf{l}_{\text{elapsed}} = \left(l_{\text{elapsed}|\text{R}},\, l_{\text{elapsed}|\text{B}}\right).$$

If these componentwise conditions hold, l_2 can be discarded because it cannot yield a better feasible completion than l_1.

The scalar full-recharge dominance rule is therefore a special case of the partial-recharge dominance rule in the main text, obtained when the time and energy intervals collapse to singletons.

Algorithm 1 Benchmark instance generation

Require: Route class r, number of clusters K, number of customers N, allocation type a, random seed
Ensure: Instance data, distance matrix, time matrix, and diagnostics

Sample the total route length D according to route class r
Place K cluster centers approximately evenly along $[0, D]$, with small random perturbations
if $K = 1$ **then**
    Assign all customers to the single cluster
else if $a =$ BALANCED **then**
    Distribute the N customers approximately evenly among the clusters
else
    Select one dominant cluster at random
    Assign most customers to the dominant cluster and only a few to each remaining cluster
end if
Sample customer positions around their assigned cluster centers
Adjust customer positions to satisfy the required spacing constraints
Construct the predetermined visit sequence from the origin through the customers to the destination
Compute pairwise route distances d_{ij}
Compute travel times as $t_{ij} \leftarrow d_{ij}/60$, assuming a constant speed of 60 km/h
Generate customer service times
if the feasibility conditions are violated **then**
    Resample the instance until a feasible configuration is obtained
end if
Write the instance file, distance and time matrices, and diagnostic data

Appendix B. Pseudocode benchmark instance generator

Appendix C. Driver-inspired sequential heuristic

Algorithm 2 Driver-inspired sequential heuristic

Require: Prescribed route, battery capacity B, charging power P_{ch}, consumption rate α, CS spacing d_{CS}, synchronization horizon t_{sync}, policy $p \in \{\textsc{Full}, \textsc{Overlap}\}$

Ensure: Feasible activity schedule or an infeasibility declaration

Insert the charging opportunities along the prescribed route
Initialize time and HoS counters to zero and SOC $\leftarrow B$
for each next route node or charging opportunity j **do**
    while j has not been reached **do**
        Set Δ to the minimum of the remaining travel time, time to the next mandatory break, and time to the next daily rest
        Drive for Δ; update time, SOC, and HoS counters
        if an HoS limit binds before reaching j **then**
            Take a daily rest if required; otherwise take a break ▷ Rule 4
        end if
    end while
    if j is a charging station **then**
        Compute energy E_{next} needed to reach the next charging opportunity or destination
        Compute times t_B and t_R until a break and daily rest become mandatory
        if SOC $< E_{\text{next}}$ **then**
            Select a daily rest if $t_R < t_{\text{sync}}$; otherwise select a break
            Mark the stop as necessary charging ▷ Rule 2
        else if $t_R < t_{\text{sync}}$ or $t_B < t_{\text{sync}}$ **then**
            Select a daily rest if $t_R < t_{\text{sync}}$; otherwise select a break
            Mark the stop as early synchronization ▷ Rule 3
        else
            Pass the charging station without stopping ▷ Rule 1
            continue
        end if
        if $p = \textsc{Full}$ **then**
            Set target SOC to B
        else
            Set target SOC to the energy obtainable during the selected interruption
            Increase target SOC to E_{next} if required
        end if
        if a break and its excess charging time violate the daily elapsed limit **then**
            Replace the break by a daily rest
        end if
        Perform charging and the selected interruption concurrently
    else
        if service at j would violate an HoS elapsed-time limit **then**
            Take the required break or daily rest
        end if
        Perform service at j
    end if
end for
return schedule if valid; otherwise declare infeasibility

Appendix D. Summary of benchmark instances

We should make this openly available on say Github. Add a link

Table D.8: Key columns only. Distances in km; times in h. Abbreviations: Test = scenario/test name; Class = route class (short/med/long); K = number of clusters; N = total customers; Var = variant (Balanced/Unbalanced); Dist_km = total route distance; Time_h = total route time; Cust/Cl = customers per cluster; AvgCl_km = average cluster distance; AvgIntra_km = average intra-cluster customer distance; AvgInter_km = average inter-cluster distance; AvgSvc_h = average service time per customer.

Test	Class	K	N	Var	Dist_km	Time_h	Cust/Cl	AvgCl_km	AvgIntra_km	AvgInter_km	AvgSvc_h
long_cl1_cu4	long	1	4	-	3929.38	56.1340	[4]	0.00	30.00	0.00	0.2650
long_cl2_cu2_Balanced	long	2	2	Balanced	4485.46	64.0780	[1,1]	674.75	0.00	674.75	0.4579
long_cl2_cu2_Unbalanced	long	2	2	Unbalanced	4472.53	63.8932	[1,1]	972.92	0.00	972.92	0.4207
long_cl2_cu3_Balanced	long	2	3	Balanced	4400.36	62.8623	[1,2]	797.97	37.73	797.97	0.4761
long_cl2_cu4_Balanced	long	2	4	Balanced	4162.88	59.4697	[2,2]	459.89	52.93	459.89	0.2409
long_cl2_cu4_Unbalanced	long	2	4	Unbalanced	4576.22	65.3745	[3,1]	1030.26	103.29	1030.26	0.2304
long_cl2_cu5_Balanced	long	2	5	Balanced	5119.54	73.1362	[3,2]	988.26	73.77	988.26	0.2437
long_cl3_cu12_Balanced	long	3	12	Balanced	5097.76	72.8252	[4,4,4]	810.85	45.65	1081.13	0.1286
long_cl3_cu12_Unbalanced	long	3	12	Unbalanced	4355.65	62.2236	[1,10,1]	647.62	30.00	863.49	0.1340
long_cl3_cu14_Balanced	long	3	14	Balanced	5063.93	72.3419	[5,5,4]	807.74	30.00	1076.99	0.1190
long_cl3_cu3_Balanced	long	3	3	Balanced	5016.56	71.6651	[1,1,1]	818.32	0.00	1091.09	0.3899
long_cl3_cu4_Balanced	long	3	4	Balanced	4155.45	59.3636	[2,1,1]	680.25	62.77	907.00	0.2582
long_cl3_cu4_Unbalanced	long	3	4	Unbalanced	5020.14	71.7163	[1,2,1]	758.57	50.84	1011.43	0.2508
long_cl3_cu5_Balanced	long	3	5	Balanced	5518.38	78.8340	[2,2,1]	894.65	44.15	1192.87	0.2550
med_cl1_cu2	med	1	2	-	2710.05	38.7150	[2]	0.00	84.81	0.00	0.4260
med_cl1_cu4	med	1	4	-	2615.02	37.3575	[4]	0.00	30.00	0.00	0.2420
med_cl2_cu2_Balanced	med	2	2	Balanced	2737.17	39.1024	[1,1]	446.85	0.00	446.85	0.4159
med_cl2_cu2_Unbalanced	med	2	2	Unbalanced	2321.67	33.1667	[1,1]	258.77	0.00	258.77	0.4043
med_cl2_cu3_Balanced	med	2	3	Balanced	3074.20	43.9171	[1,2]	499.00	43.26	499.00	0.4393
med_cl2_cu4_Balanced	med	2	4	Balanced	2413.01	34.4716	[2,2]	436.35	42.90	436.35	0.2751
med_cl2_cu4_Unbalanced	med	2	4	Unbalanced	3105.84	44.3691	[2,2]	681.02	73.82	681.02	0.2348
med_cl2_cu5_Balanced	med	2	5	Balanced	2251.60	32.1658	[3,2]	286.03	47.68	286.03	0.2607
med_cl3_cu11_Balanced	med	3	11	Balanced	2534.30	36.2043	[4,4,3]	332.33	56.27	443.11	0.1281
med_cl3_cu13_Balanced	med	3	13	Balanced	2513.67	35.9096	[5,4,4]	369.80	30.00	493.07	0.1271
med_cl3_cu13_Unbalanced	med	3	13	Unbalanced	3189.23	45.5604	[2,2,9]	437.50	36.03	583.33	0.1135
med_cl3_cu3_Balanced	med	3	3	Balanced	3292.08	47.0297	[1,1,1]	477.29	0.00	636.38	0.4144
med_cl3_cu4_Balanced	med	3	4	Balanced	2265.07	32.3581	[2,1,1]	360.47	142.64	480.63	0.2684
med_cl3_cu4_Unbalanced	med	3	4	Unbalanced	2200.53	31.4361	[2,1,1]	399.06	132.66	532.08	0.2710
med_cl3_cu5_Balanced	med	3	5	Balanced	3472.59	49.6084	[2,2,1]	559.40	34.92	745.87	0.2871
short_cl1_cu2	short	1	2	-	1081.84	15.4549	[2]	0.00	39.79	0.00	0.4019
short_cl2_cu2_Balanced	short	2	2	Balanced	1306.92	18.6703	[1,1]	213.84	0.00	213.84	0.4766
short_cl2_cu2_Unbalanced	short	2	2	Unbalanced	1512.53	21.6076	[1,1]	221.86	0.00	221.86	0.4272
short_cl2_cu3_Balanced	short	2	3	Balanced	1109.01	15.8429	[1,2]	108.55	72.96	108.55	0.3558
short_cl2_cu4_Balanced	short	2	4	Balanced	1221.36	17.4480	[2,2]	162.51	84.56	162.51	0.2685
short_cl2_cu4_Unbalanced	short	2	4	Unbalanced	1408.63	20.1233	[1,3]	159.19	93.35	159.19	0.2364
short_cl2_cu5_Balanced	short	2	5	Balanced	1249.49	17.8499	[3,2]	175.63	35.51	175.63	0.2610
short_cl3_cu10_Balanced	short	3	10	Balanced	1761.42	25.1631	[4,3,3]	260.70	30.00	347.60	0.1333
short_cl3_cu12_Balanced	short	3	12	Balanced	1941.81	27.7401	[4,4,4]	277.09	43.40	369.45	0.1206
short_cl3_cu12_Unbalanced	short	3	12	Unbalanced	1541.88	22.0269	[8,2,2]	215.62	43.30	287.50	0.1268
short_cl3_cu3_Balanced	short	3	3	Balanced	1450.69	20.7242	[1,1,1]	234.78	0.00	313.04	0.4172
short_cl3_cu4_Balanced	short	3	4	Balanced	1270.79	18.1542	[2,1,1]	241.75	106.66	322.33	0.2682
short_cl3_cu4_Unbalanced	short	3	4	Unbalanced	1324.85	18.9264	[2,1,1]	197.50	31.86	263.33	0.2117

short_cl3_cu5_Balanced	short	3	5	Balanced	1876.99	26.8142	[2,2,1]	259.19	68.40	345.58	0.2621

Appendix E. Additional Per-Instance Comparison

Table E.9: Near-diesel diagnostic cases in the base-case setting.

Instance	EV (h)	Diesel (h)	CS	Sync. (%)	Unsync.	Idle (min)
s_cl2_cu4_U	27.86	27.86	3	100.0	0	0
s_cl2_cu3_B	27.19	27.19	2	100.0	0	0
m_cl2_cu2_U	68.95	68.95	6	100.0	0	0
m_cl3_cu4_U	64.10	64.10	5	100.0	0	0
s_cl3_cu10_B	44.20	44.20	4	100.0	0	0
s_cl2_cu4_B	44.19	44.19	2	100.0	0	0

These are the base-case instances for which the optimized electric schedule attains the same total route time as the diesel proxy. In all six cases, every charging stop is synchronized with a mandatory break or rest.

Table E.10: Full- and partial-recharge results from the computational experiments. Times are in hours. Abbreviations: ID = short instance identifier used in the results files; Test = full instance name; F_Time = full-recharge route time; F_CS = number of charging stops under full recharge; F_Sync = number of synchronized charging stops under full recharge; F_Unsync = number of unsynchronized charging stops under full recharge; F_Break = number of full-recharge charging stops synchronized with breaks; F_Rest = number of full-recharge charging stops synchronized with daily rests; P_Time = partial-recharge route time; P_CS = number of charging stops under partial recharge; P_Sync = number of synchronized charging stops under partial recharge; P_Unsync = number of unsynchronized charging stops under partial recharge; P_Break = number of partial-recharge charging stops synchronized with breaks; P_Rest = number of partial-recharge charging stops synchronized with daily rests.

ID	Test	F_Time	F_CS	F_Sync	F_Unsync	F_Break	F_Rest	P_Time	P_CS	P_Sync	P_Unsync	P_Break	P_Rest
S_cl1_cu2_B	short_cl1_cu2	31.73	3	1	2	0	1	31.49	3	1	2	0	1
S_cl2_cu3_B	short_cl2_cu3_Balanced	27.27	1	1	0	0	1	27.19	2	2	0	1	1
S_cl2_cu4_B	short_cl2_cu4_Balanced	44.35	2	2	0	0	2	44.19	2	2	0	0	2
S_cl2_cu4_U	short_cl2_cu4_Unbalanced	27.94	1	1	0	0	1	27.86	3	3	0	2	1
S_cl2_cu5_B	short_cl2_cu5_Balanced	30.88	2	1	1	0	1	30.72	2	1	1	0	1
S_cl3_cu10_B	short_cl3_cu10_Balanced	44.36	2	2	0	0	2	44.20	4	4	0	2	2
S_cl3_cu12_B	short_cl3_cu12_Balanced	43.69	2	2	0	0	2	43.53	4	4	0	2	2
S_cl3_cu12_U	short_cl3_cu12_Unbalanced	30.57	2	1	1	0	1	30.16	2	2	0	1	1
S_cl3_cu3_B	short_cl3_cu3_Balanced	31.35	3	1	2	0	1	30.86	3	2	1	1	1
S_cl3_cu4_B	short_cl3_cu4_Balanced	31.18	2	1	1	0	1	31.02	3	2	1	1	1
S_cl3_cu4_U	short_cl3_cu4_Unbalanced	32.17	2	1	1	0	1	31.85	3	2	1	1	1
S_cl3_cu5_B	short_cl3_cu5_Balanced	31.48	2	1	1	0	1	31.32	3	2	1	1	1
M_cl1_cu2_B	med_cl1_cu2	68.95	6	6	0	0	6	68.95	6	6	0	0	6
M_cl1_cu4_B	med_cl1_cu4	64.26	5	4	1	0	4	64.10	5	5	0	1	4
M_cl2_cu2_B	med_cl2_cu2_Balanced	69.44	4	3	1	0	3	68.95	6	6	0	0	6
M_cl2_cu2_U	med_cl2_cu2_Unbalanced	37.11	2	2	0	0	2	36.62	3	3	0	1	2
M_cl2_cu3_B	med_cl2_cu3_Balanced	47.95	3	3	0	0	3	47.46	4	4	0	1	3
M_cl2_cu4_B	med_cl2_cu4_Balanced	64.09	5	5	0	0	5	63.60	7	7	0	2	5
M_cl2_cu4_U	med_cl2_cu4_Unbalanced	65.70	4	3	1	0	3	65.20	6	5	1	2	3
M_cl2_cu5_B	med_cl2_cu5_Balanced	37.37	2	2	0	0	2	36.88	3	3	0	1	2
M_cl3_cu11_B	med_cl3_cu11_Balanced	47.45	2	2	0	0	2	47.29	4	4	0	2	2
M_cl3_cu13_B	med_cl3_cu13_Balanced	48.39	2	2	0	0	2	48.23	4	4	0	2	2
M_cl3_cu13_U	med_cl3_cu13_Unbalanced	48.98	2	2	0	0	2	48.82	4	4	0	2	2
M_cl3_cu3_B	med_cl3_cu3_Balanced	48.66	4	3	1	0	3	47.84	5	4	1	1	3
M_cl3_cu4_B	med_cl3_cu4_Balanced	38.37	2	2	0	0	2	37.88	3	3	0	1	2

M_cl3_cu4_U	med_cl3_cu4_Unbalanced	33.00	2	2	0	0	2	32.51	3	3	0	1	2
M_cl3_cu5_B	med_cl3_cu5_Balanced	50.26	3	2	1	0	2	49.77	4	3	1	1	2
L_cl1_cu2_B	long_cl1_cu2	74.61	6	4	2	0	4	74.12	7	5	2	1	4
L_cl1_cu4_B	long_cl1_cu4	86.24	5	3	2	0	3	85.75	7	5	2	2	3
L_cl2_cu2_B	long_cl2_cu2_Balanced	75.95	4	3	1	0	3	74.65	7	6	1	2	4
L_cl2_cu2_U	long_cl2_cu2_Unbalanced	76.55	4	3	1	0	3	75.06	6	5	1	2	3
L_cl2_cu3_B	long_cl2_cu3_Balanced	76.02	5	3	2	0	3	74.53	7	5	2	2	3
L_cl2_cu4_B	long_cl2_cu4_Balanced	65.36	4	3	1	0	3	64.05	5	4	1	1	3
L_cl2_cu4_U	long_cl2_cu4_Unbalanced	79.11	6	3	3	0	3	77.81	8	5	3	2	3
L_cl2_cu5_B	long_cl2_cu5_Balanced	88.17	5	3	2	0	3	87.68	7	5	2	2	3
L_cl3_cu12_B	long_cl3_cu12_Balanced	74.79	4	4	0	0	4	74.63	8	8	0	4	4
L_cl3_cu12_U	long_cl3_cu12_Unbalanced	87.07	5	3	2	0	3	86.58	6	4	2	1	3
L_cl3_cu14_B	long_cl3_cu14_Balanced	73.33	4	4	0	0	4	73.16	8	8	0	4	4
L_cl3_cu3_B	long_cl3_cu3_Balanced	88.04	6	4	2	0	4	86.75	7	5	2	1	4
L_cl3_cu4_B	long_cl3_cu4_Balanced	76.94	5	3	2	0	3	75.45	7	5	2	2	3
L_cl3_cu4_U	long_cl3_cu4_Unbalanced	85.90	7	4	3	0	4	84.41	9	6	3	2	4
L_cl3_cu5_B	long_cl3_cu5_Balanced	98.45	6	4	2	0	4	97.96	8	6	2	2	4